# Social Curiosity: Study 3 (Forced-Choice) (#209319)

**Authors**
This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

**Pre-registered on:**
2025/01/23 11:31 (PT)

**1) Have any data been collected for this study already?**
No, no data have been collected for this study yet.

**2) What's the main question being asked or hypothesis being tested in this study?**
Do preschool-aged children seek out information about how others evaluated them? We hypothesize that 4- and 5-year-old children are more curious to find out how others evaluated their own work versus the work of another child. Specifically, we predict that children would be more likely to look at their own evaluation (rather than another individual's) when given explicit permission to look at either one.

**3) Describe the key dependent variable(s) specifying how they will be measured.**
After an experimenter evaluates the participant's and another child's (i.e., "Jordan", an unfamiliar, gender-matched child to the participant) tracings out of sight, they will give children permission to look at one of the evaluations. The key dependent measure is which folder children choose to look inside of.

**4) How many and which conditions will participants be assigned to?**
Single condition experiment. Children will make a tracing (tree or bird; counterbalanced) then be shown another unfamiliar child's tracing (tree or bird, whichever the child did not trace). The two tracings will be placed inside of separate folders and evaluated out of sight (order of evaluation and folder color counterbalanced).

After evaluation, the experimenter will offer for children to look inside of one of the folders to see the evaluation.

**5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.**
To analyze whether participants' choice of folder differs from chance, we will compute the Bayes Factor test of a single proportion, using the proportionBF command from the BayesFactor package in R (Morey et al., 2024).

i.e., proportionBF(k, n, p = 0.5)

We will use the package default priors: a point distribution at 0.5 representing H0, and a logistic distribution centered at 0 (i.e., log-odds corresponding to 0.5) with the default prior r scale ½ representing H1.

We will interpret a BF10 > 10 as strong evidence, > 5 as moderate evidence, and > 3 as weak but meaningful evidence in favor of H1. We will use the same thresholds to interpret evidence in favor of the null (i.e., BF01)

**6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.**
Children will be excluded for interference (e.g., due to teachers, children, or other individuals outside the research room), experimenter error (e.g., large deviations from the study script), looking at one of the folders prior to the key test phase, failing to complete the study, having parent-reported developmental delays, or technical errors (e.g., video does not record).

**7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.**
We are uncertain as to the size of the effect we will observe (if any), thus we plan a sequential Bayes Factor analysis (e.g., Mani et al., 2021). We will calculate the Bayes Factor via the analysis plan outlined above.

We will test an initial sample of 20 children (4- and 5-year-olds), then evaluate the Bayes Factor (i.e., BF10) after every 4 data points. We will stop testing at BF10 > 10, BF10 < ⅕, or at an N = 36.

**8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)**
For consistency with previous studies in this line of work, we will also conduct a frequentist analysis of the data (Exact Binomial Test) at the conclusion of data collection.